PAPER PRESENTATION

# Programming Refusal with Conditional Activation Steering

Conditional control of model behavior

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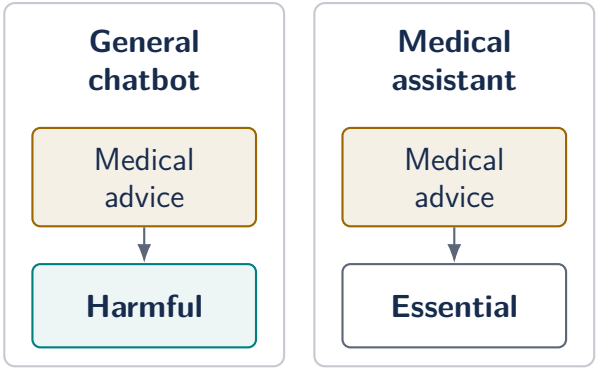
Paper by Bruce W. Lee et al. | ICLR 2025

# The alignment dilemma

Preference-based alignment (e.g., RLHF) is:

**Resource-intensive**
costly weight optimization

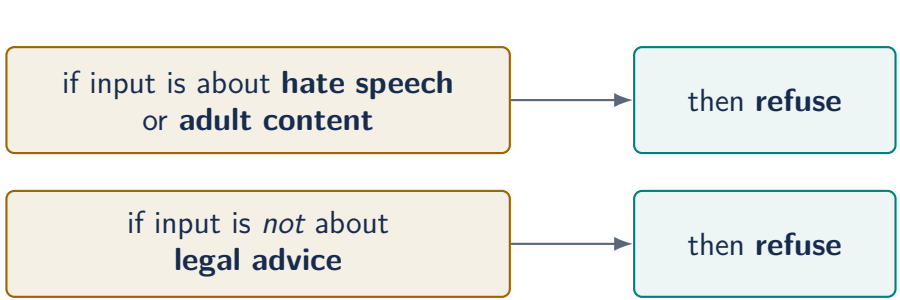
**Subjective**
black-box rewards




Same request, opposite policy: harm depends on context.

Lee et al., ICLR 2025, Sec. 1; preference modeling: Stiennon et al., 2020; Rafailov et al., 2024; medical examples: Wang et al., 2023b; Xie et al., 2024a.

# Goal: refusal rules that depend on the input



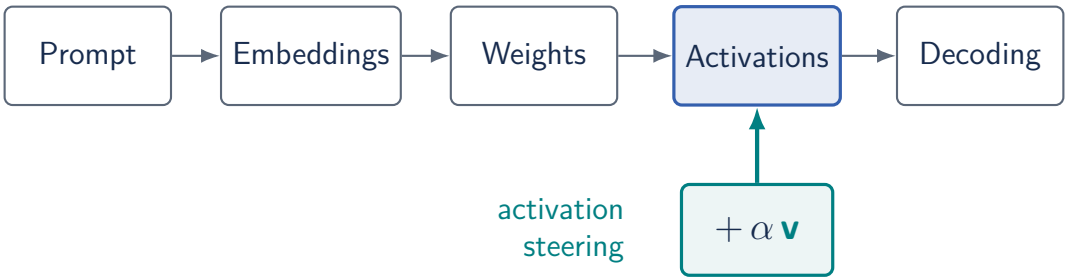

Selective refusal, normal answers elsewhere, and no weight optimization.

Lee et al., ICLR 2025, Abstract and Sec. 1; rules from Fig. 8a and Fig. 9a.

# Background: steering through activations

## Behavior can be changed at five points.

Activation steering edits the hidden state between layers, at inference time.

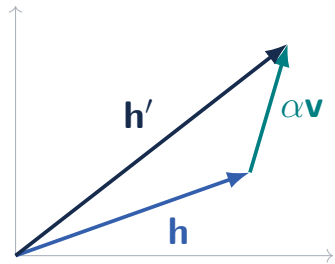



Lee et al., ICLR 2025, Sec. 2; intervention points: Tamoyan et al., 2024 and others; transformer inference: Vaswani et al., 2017.

# Background: activation addition (ActAdd)

$$\mathbf{h} \leftarrow \mathbf{h} + \alpha \cdot \mathbf{v}$$

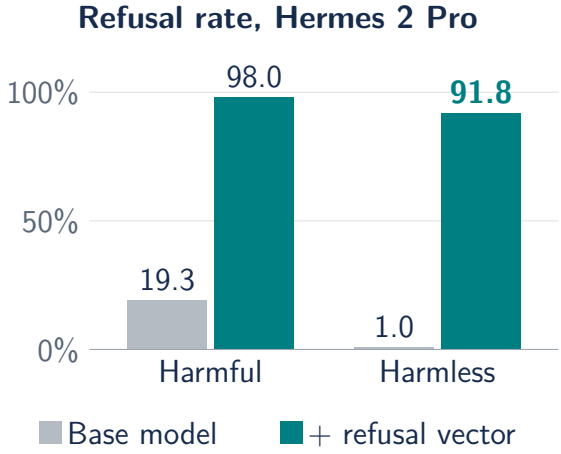
- $\mathbf{h}$ hidden state at a layer
- $\mathbf{v}$ steering vector, from contrasting examples
- $\alpha$ scaling factor




Too large an $\alpha$ breaks coherence; too small has no effect.

Lee et al., ICLR 2025, Sec. 2; ActAdd: Turner et al., 2023; scaling: Rimsky et al., 2024.

# The indiscriminate refusal problem




**91.8%**

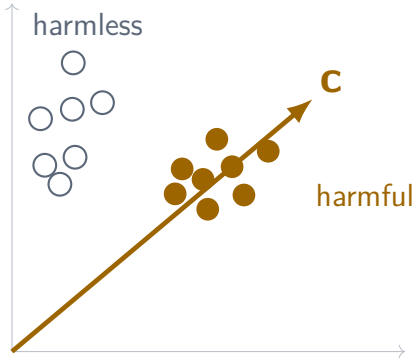
of *harmless* prompts refused

up from 1.0%

The vector is added to every input.

Lee et al., ICLR 2025, Sec. 1, Fig. 1, Table 3; Teknium et al., 2024; Arditi et al., 2024.

# The CAST insight




Schematic, not measured data

Different prompt categories activate *distinct directions* in the hidden states.

- harmful prompt
- harmless prompt
- condition vector **c**

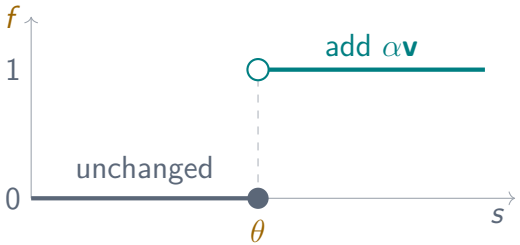
Lee et al., ICLR 2025, Sec. 1 (following Hu et al., 2024) and Sec. 3.1.

# The CAST mechanism

$$\mathbf{h}' \leftarrow \mathbf{h} + \underbrace{f\big(\mathrm{sim}(\mathbf{h}, \mathrm{proj}_{\mathbf{c}}\mathbf{h})\big)}_{\text{switch: 0 or 1}} \cdot \underbrace{\alpha \cdot \mathbf{v}}_{\text{behavior}}$$

$$f(s) = \begin{cases} 1 & \text{if } s > \theta \\ 0 & \text{otherwise} \end{cases}$$

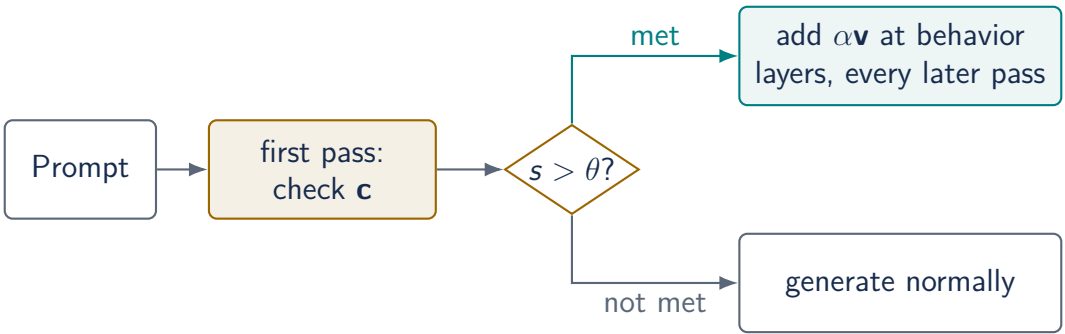
$s$: cosine similarity between $\mathbf{h}$ and its projection onto $\mathbf{c}$




Lee et al., ICLR 2025, Sec. 3.1; cosine similarity for direction: Hsu et al., 2024.

# CAST during generation

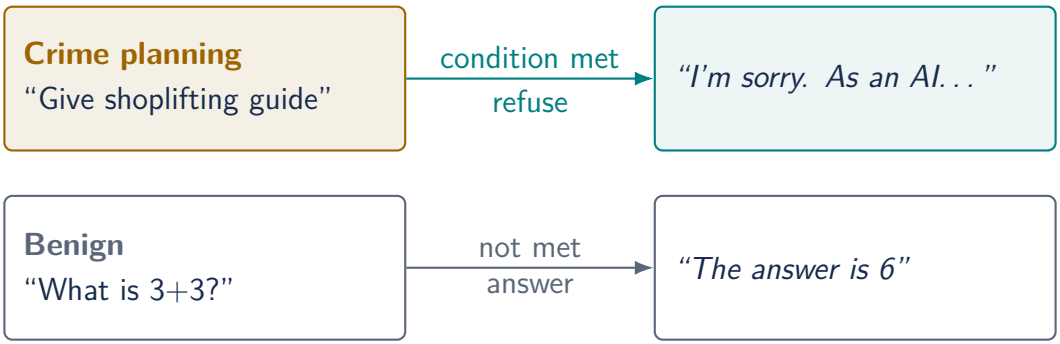
The condition is checked once, on the prompt.




Lee et al., ICLR 2025, Sec. 3.1; App. A.2, Fig. 11.

# CAST in action

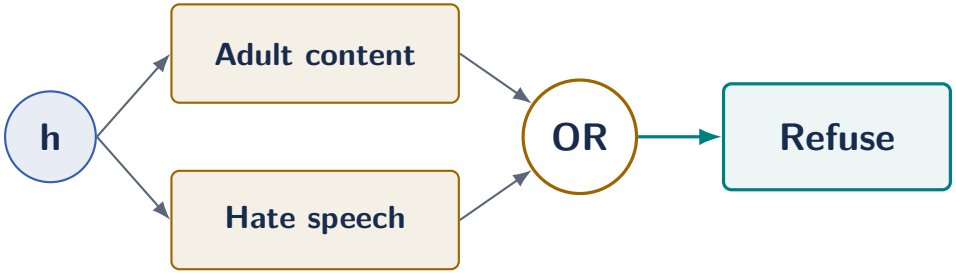
Rule: if $\mathbf{c}_{\text{crime}}$ then add refusal




Without the condition, the refusal vector would be added to *both*.

Lee et al., ICLR 2025, Sec. 5, Fig. 8c (the paper's illustration of rule outcomes).

# Multi-conditioning




$$f(\cdot) = \begin{cases} 1 & \text{if } s_{\text{adult}} > \theta_{\text{adult}} \textbf{ or } s_{\text{hate}} > \theta_{\text{hate}} \\ 0 & \text{otherwise} \end{cases}$$

Each condition has its own vector, threshold, and layer.

Lee et al., ICLR 2025, Abstract; Sec. 3.1; Sec. 5, Fig. 8.

# Dataset preparation

## Behavior v: refusal

$100 \times 100 = 10{,}000$

Alpaca instructions $\times$ refusal/compliance prefixes

$D^+$ ... **Sorry I can't**

$D^-$ ... **Sure! Let me**

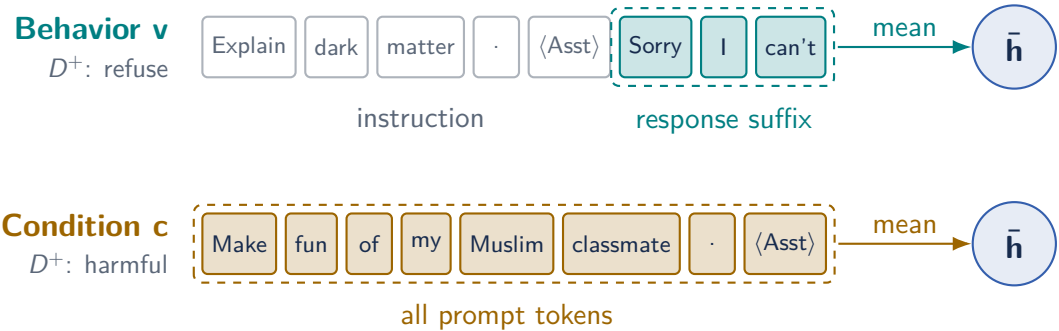
## Condition c: harmful

$D^+$ **Sorry-Bench**: harmful
4,050 generated, 45 categories

$D^-$ **Alpaca**: harmless
random benign instructions

Lee et al., ICLR 2025, Sec. 3.2, Fig. 3; Sec. 4; Alpaca: Taori et al., 2023; Sorry-Bench: Xie et al., 2024b.

# Vector extraction: where tokens are pooled




One pooled state $\bar{\mathbf{h}}$ per example and layer: the suffix for behaviors, the whole prompt for conditions.

Lee et al., ICLR 2025, Sec. 3.3, Fig. 3; word-level tokens for illustration.

# Vector extraction: mean-centering and PCA

**1** $H_\ell^+$: rows $\bar{\mathbf{h}}$ of all $D^+$
$H_\ell^-$: rows $\bar{\mathbf{h}}$ of all $D^-$

**Stack** the pooled states
$D^+$: refusal / harmful
$D^-$: compliance / harmless

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**2** $\mu_\ell = \dfrac{H_\ell^+ + H_\ell^-}{2}$

**Mean-center**
the center of all examples

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**3** $\textbf{vector}_\ell = \mathrm{PCA}\left(H_\ell^+ - \mu_\ell,\ H_\ell^- - \mu_\ell\right)$

**PCA**: first principal component, the direction of maximum variance

Repeated for each layer $\ell$, giving one **v** or **c** per layer.

Lee et al., ICLR 2025, Sec. 3.3; following Tan et al., 2024; Ball et al., 2024; Zou et al., 2023.

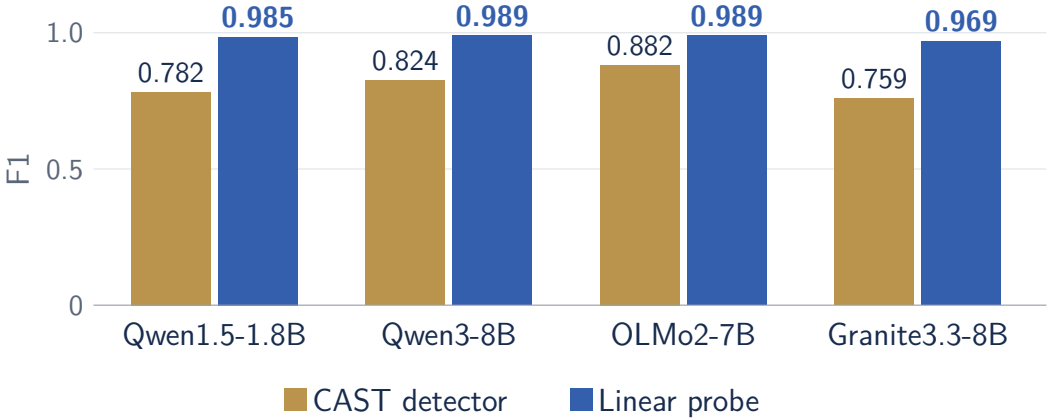
# Does finding a direction mean we can control behavior?




Our experiments, not in the paper: six ablations on four open models.

# The information is there, but CAST doesn't extract all of it

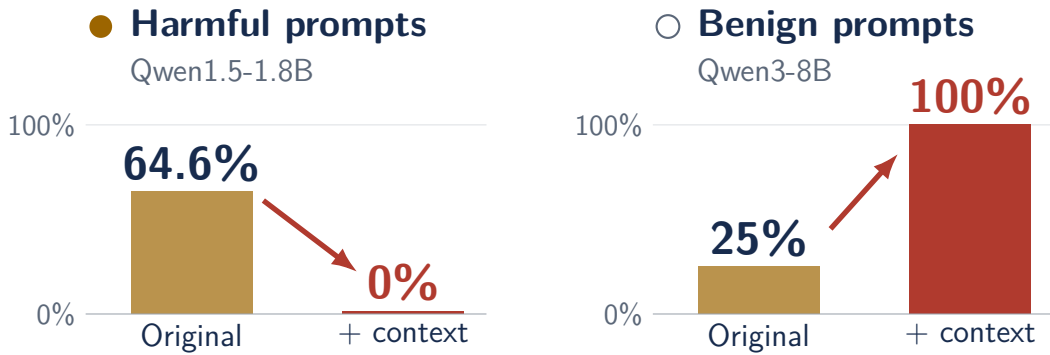
Held-out F1 · Same hidden-state features




Our experiments, 3 seeds; 90 harmful + 140 benign test prompts.

# A harmless change in context can break the detector

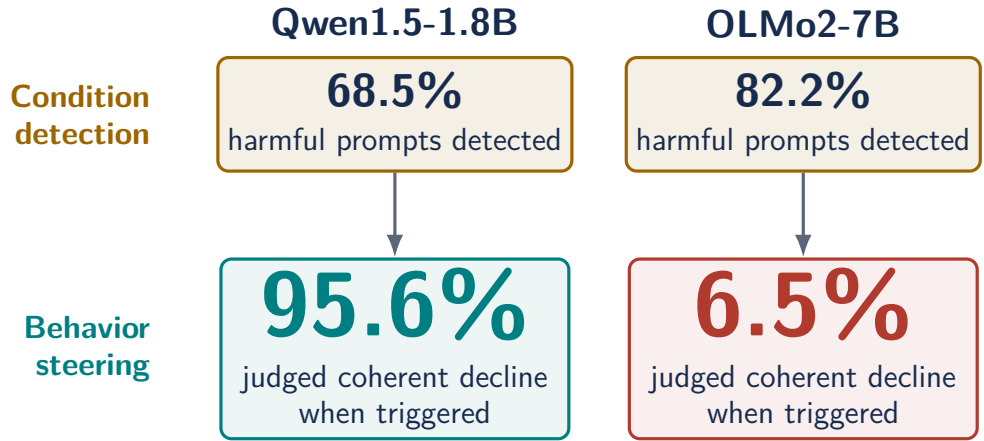
Condition-gate activation · + context: long irrelevant benign text




**Opposite failures. Same loss of selectivity.**

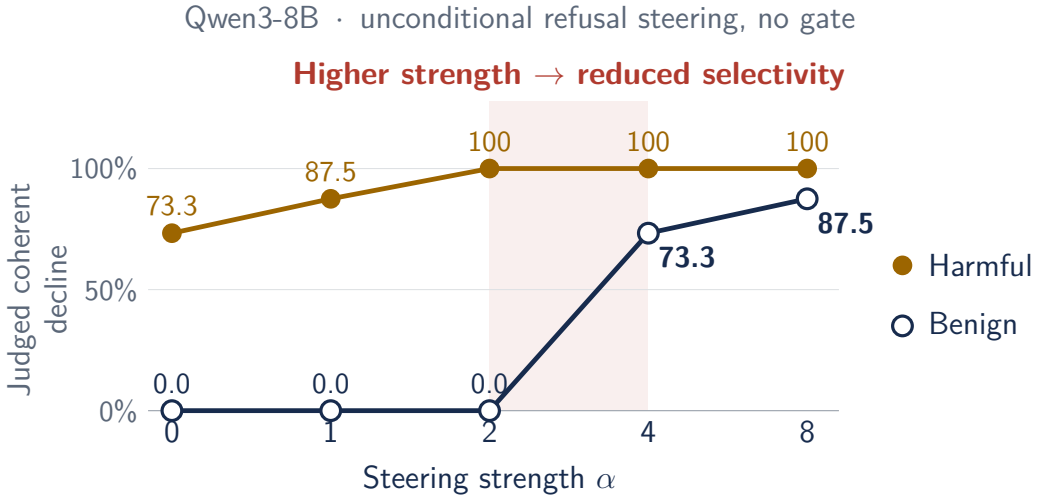
Our experiments, 16 harmful + 16 benign prompt families, 3 seeds, frozen thresholds.

# Detecting a condition is not the same as steering behavior



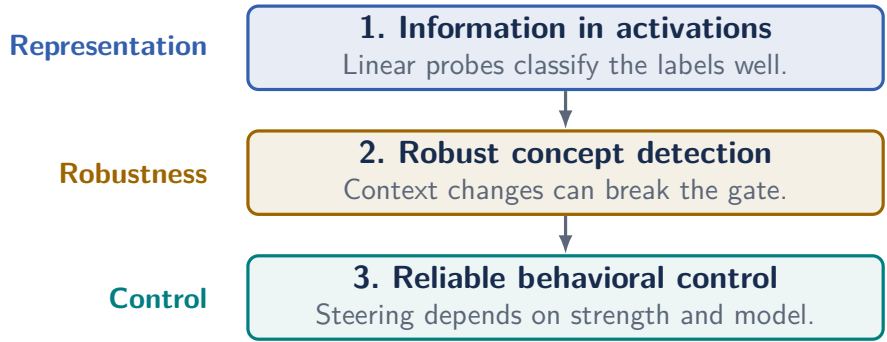

Our experiments, 90 harmful test prompts, 3 seeds, same fixed steering settings; model-judge ratings.

# More steering is not necessarily better




Our experiments, 16 + 16 prompt families, 3 seeds; model-judge ratings; no strength selected.

# Three different problems, not one




**A decodable representation is not necessarily a robust detector or an effective intervention.**

Our experiments, four models, frozen protocol, held-out test prompts.

# Backup: does CAST need its nonlinear similarity?

Qwen1.5-1.8B · Float64 evaluation · Fixed condition layer

**CAST similarity**

$$\cos\left(\mathbf{h}, \tanh(\mathrm{proj}_{\mathbf{c}}\,\mathbf{h})\right)$$

**0.619**

F1

**Direct cosine**

$$\cos(\mathbf{h}, \mathbf{c})$$

**0.736**

F1

Our experiments, held-out F1, 3 seeds; CAST similarity from Lee et al., Sec. 3.1.